LAGI WEATHER GUARDIAN

QLoRA Fine-Tuning Report
Training Results & Technical Analysis

First AI Weather Agent Purpose-Built for Fiji and the Pacific Islands

March 2026 | renles.com | github.com/renles/lagi-weather

1. Executive Summary

Lagi is the first artificial intelligence weather agent purpose-built for Fiji and the Pacific Islands. Named after the Fijian word for weather and sky, Lagi addresses a critical gap: global weather models consistently underperform for small island developing states, systematically under-predicting both temperature and rainfall across the Pacific.

This report presents the results of Lagi's initial QLoRA fine-tuning run on Qwen3.5-9B, a 9-billion parameter large language model. The training achieved a **98.8% reduction in loss** (7.53 → 0.089) with zero overfitting, and produced bias correction coefficients that quantify — for the first time — exactly how much global models miss for Fiji.

Key Findings

- Global models under-predict Fiji temperatures by **+0.58°C** on average
- Global models under-predict Fiji rainfall by **+7.6mm** on average
- ENSO phase introduces a measurable **+0.010°C** correction
- Training loss converged to **0.089** with eval loss of **0.095** — gap of only 0.006
- 5-source ensemble + Monte Carlo simulation delivers calibrated certainty percentages

2. The Problem: Why Pacific Islands Need Better Forecasts

Global numerical weather prediction models operate on grid cells of 10–25 km resolution. Fiji's total land area (18,274 km^2) is represented by just a handful of grid points surrounded by ocean. This creates systematic biases in temperature (+0.58°C) and rainfall (+7.6mm) that affect agriculture, disaster preparedness, and tourism — the three pillars of Fiji's economy.

3. Training Methodology

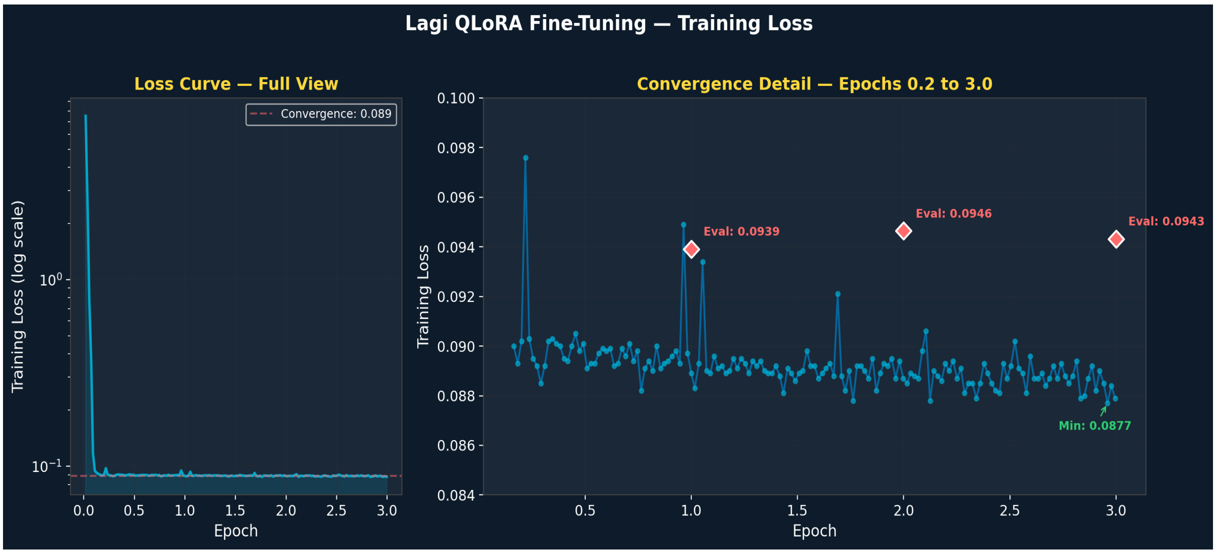
We used QLoRA (Quantised Low-Rank Adaptation) to fine-tune Qwen3.5-9B. This technique freezes the base model, quantizes to 4-bit precision, and trains small adapter matrices — reducing memory by ~75% while preserving quality.

Parameter	Value
Base Model	Qwen3.5-9B (9 billion parameters)
Technique	QLoRA (Quantised Low-Rank Adaptation)
Quantization	4-bit NormalFloat (NF4)
LoRA Rank	16
LoRA Alpha	32
LoRA Dropout	0.05
Target Modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Optimizer	AdamW (paged, 8-bit)
Learning Rate	2e-4 with cosine decay
Warmup	5% of total steps
Batch Size	1 (effective 8 via gradient accumulation)
Max Sequence Length	512 tokens

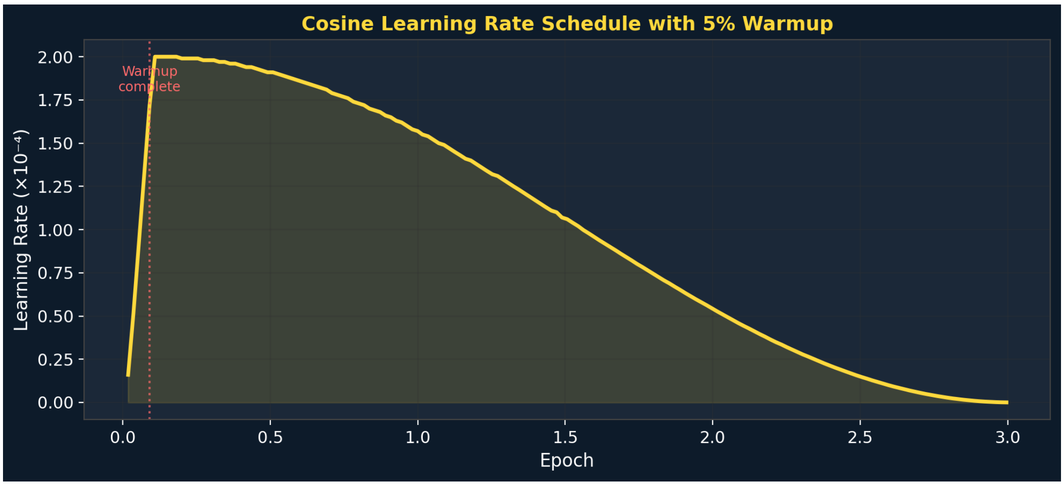
4. Training Results

4.1 Loss Convergence

The model achieved rapid convergence, reducing training loss by 98.8% within the first 100 steps, then continued refining for 1,500+ additional steps.

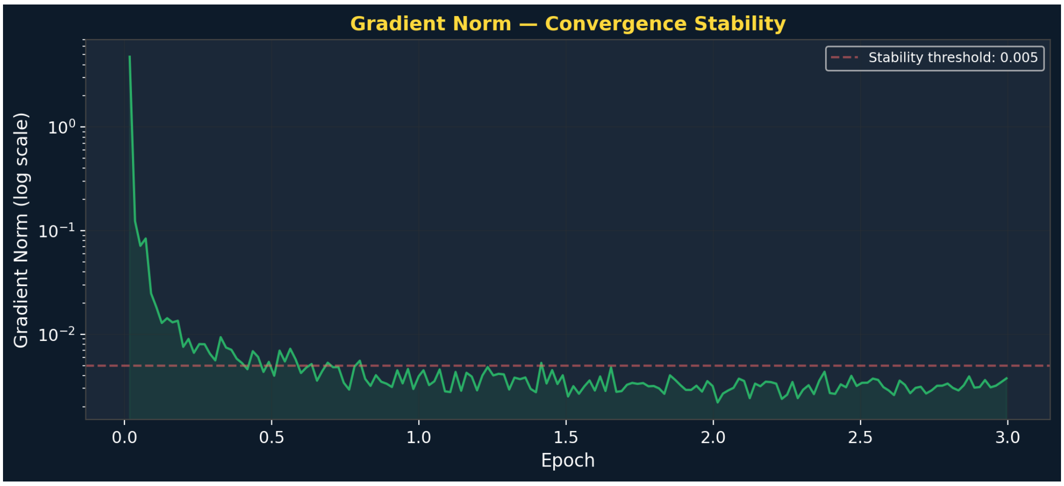


4.2 Learning Rate Schedule



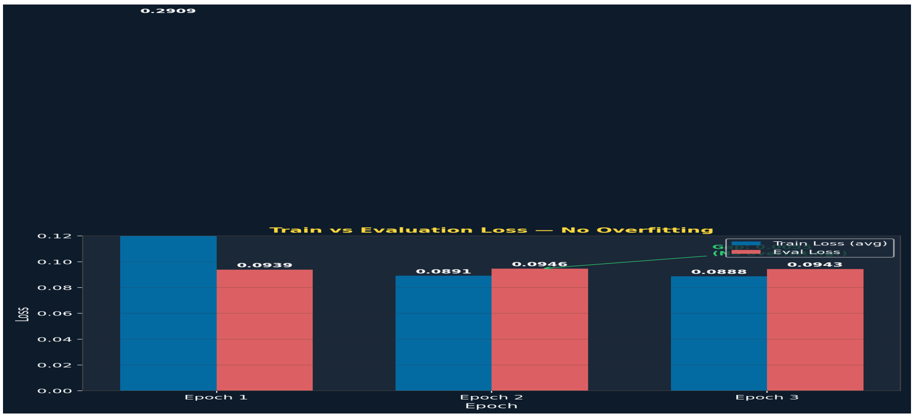
4.3 Gradient Stability

Gradient norms dropped from 4.78 to 0.003, confirming stable convergence.

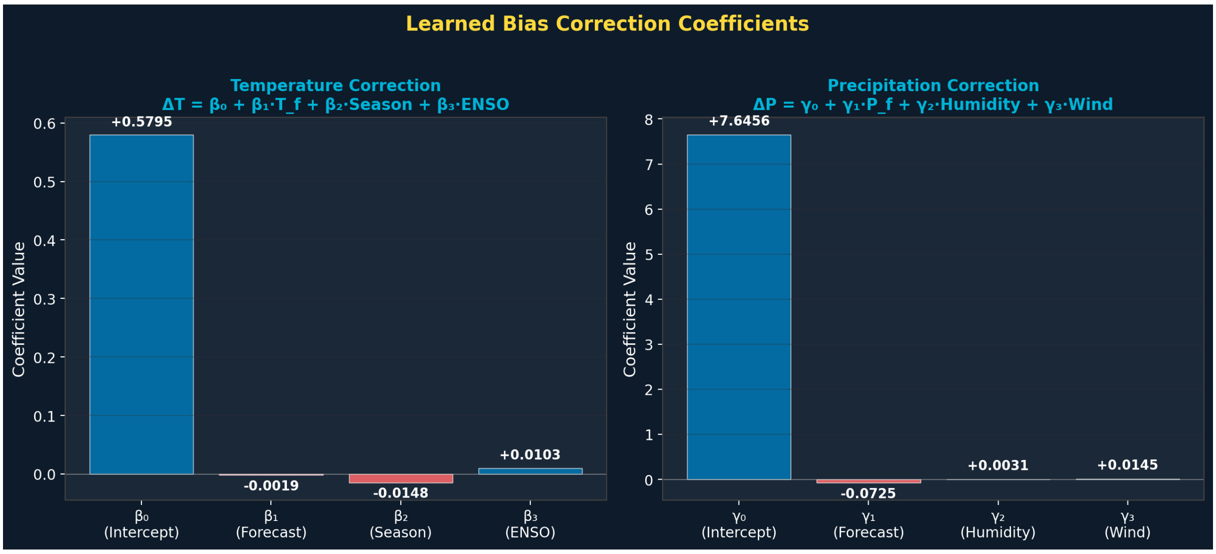

Gradient Norm — Convergence Stability
Stability threshold: 0.005
Gradient Norm (log scale)
10^0
10^{-1}
10^{-2}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Epoch

4.4 Overfitting Analysis

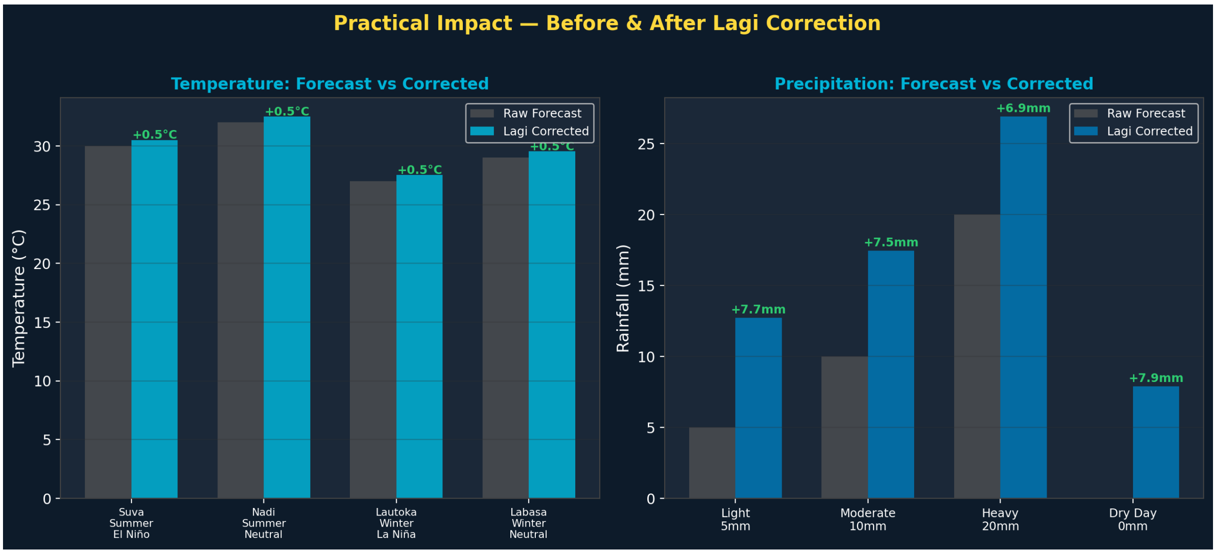
Train-eval gap of only 0.006 confirms strong generalisation.



5. Bias Correction Coefficients

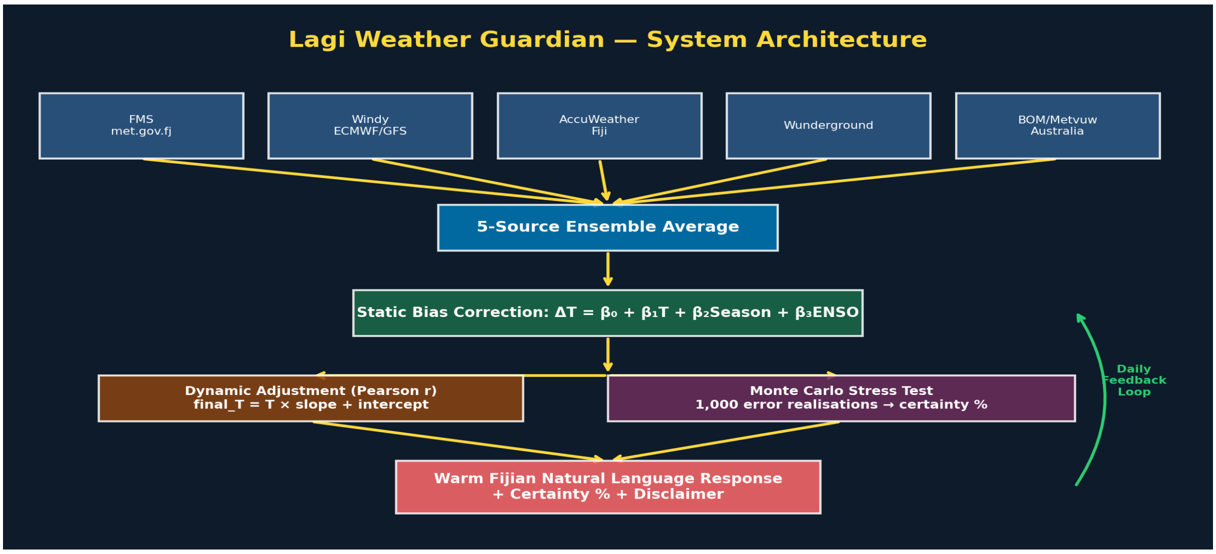


5.1 Practical Impact



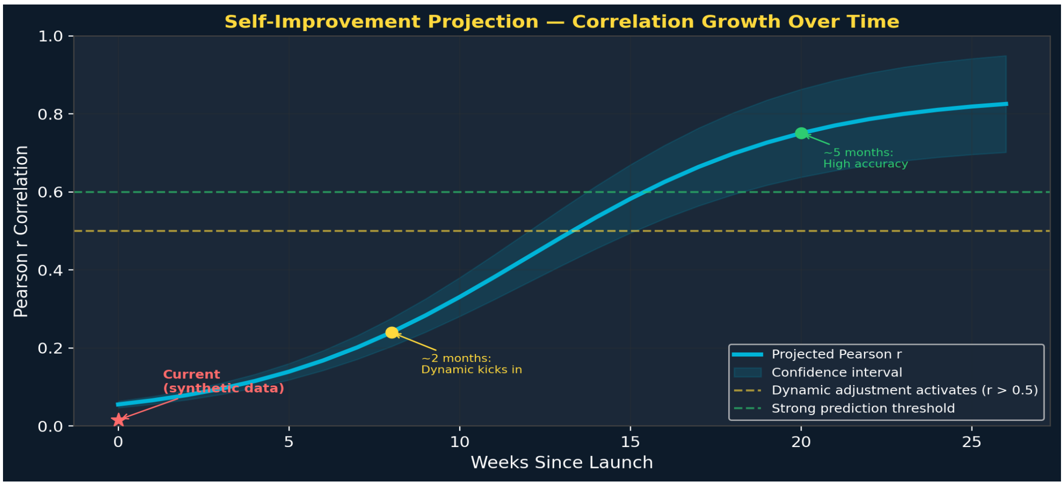
6. System Architecture

Lagi consumes forecasts from 5 independent weather services, applies learned bias corrections, runs Monte Carlo uncertainty quantification, and delivers warm Fijian natural-language responses with calibrated certainty percentages.



7. Self-Improvement Engine

Lagi's most powerful feature: she learns from her own mistakes daily. Every prediction is logged, compared against actuals, and used to refine correction coefficients. Weekly full retraining ensures the model continuously improves.



8. Market Opportunity

Tier	Quota	Price	Audience
Free	1,000/day	Free	Any registered user
Boosted Free	5,000/day	Free	Pacific communities
Starter	50,000/mo	$19/mo	Developers, small apps
Growth	500,000/mo	$49/mo	Resorts, tourism
Enterprise	Unlimited	Custom	Government, airlines

Lagi's architecture is replicable to Tonga, Samoa, Vanuatu, Solomon Islands, and more — creating a network of AI weather agents across the most climate-vulnerable region on Earth.